We have seen that if we neglect the repulsion $\frac{e^2}{r_{12}}$ between the two electrons in the ground state of He, the energy is $-8R_y = -108.8$ eV. Treating $\frac{e^2}{r_{12}}$ as a perturbation show that

$$\langle 100, 100 | H' | 100, 100 \rangle = \frac{5}{2} R_y$$

So that $E_0^0 + E_0^1 = -5.5 R_y = -74.8$ eV.

(Hint: $\langle H' \rangle$ can be viewed as the interaction between two concentric, spherically symmetric exponentially falling charge distributions. Find the potential $\varphi(r)$ due to one distribution and calculate the interaction energy between this potential and the other charge distribution)

Let us first note some definitions that we will employ.

Let $\Lambda_0 \equiv \frac{-e^2}{8\pi\varepsilon_0 a_0} = -13.6\text{eV} = -1R_y$ be the ground state energy of hydrogen. E_0^0 is the aforementioned ground state energy of helium ($-8R_y = -108.8$eV), and E_0^1 is the first order correction to the ground state energy of helium. Our Hamiltonian is

$$H = (H^0) + (H') = \left(\frac{\vec{p}_1^{\,2}}{2m} + \frac{\vec{p}_2^{\,2}}{2m} - \frac{Ze^2}{4\pi\varepsilon_0 r_1} - \frac{Ze^2}{4\pi\varepsilon_0 r_2} \right) + \left(\frac{e^2}{4\pi\varepsilon_0 r_{12}} \right) \quad (1)$$

with $Z=2$, $r_1 = |\vec{x}_1|$, $r_2 = |\vec{x}_2|$, $r_{12} = |\vec{x}_1 - \vec{x}_2|$.

Now, consider a Hamiltonian $H = H^0$ ((1) w/out H'). In this case, we may form the spatial portion of our wavefunction as products of hydrogen wavefunctions with $Z=2$ (assume nucleus stationary). In general, where $\psi_{n\ell m}$ is our hydrogen spatial wavefunction, our helium wave function will be

$$\varphi(\vec{x}_1, \vec{x}_2) = \frac{1}{\sqrt{2}} \left(\psi_{n\ell m}(\vec{x}_1) \psi_{n'\ell'm'}(\vec{x}_2) \pm \psi_{n\ell m}(\vec{x}_2) \psi_{n'\ell'm'}(\vec{x}_1) \right) \quad (2)$$

However, in the case of the state we are interested in ($n' = n = 1$; $\ell' = \ell = m' = m = 0$), (2) becomes

$$\varphi(\vec{x}_1, \vec{x}_2) = \psi_{100}(\vec{x}_1) \psi_{100}(\vec{x}_2) = \frac{Z^3}{\pi a_0^3} \exp\left(\frac{-Z(r_1 + r_2)}{a_0} \right) \quad (3)$$

where a_0 is the Bohr radius, and $\psi_{100}(\vec{x}) = \frac{1}{\sqrt{4\pi}} \left(\frac{Z}{a_0} \right)^{3/2} 2 \exp\left(-Z r/a_0 \right)$.

Now, let us introduce our perturbation, H', and use $\varphi(\vec{x}_1, \vec{x}_2) = |100; 100\rangle$ to determine the first order correction to the ground state energy of helium using perturbation theory:

$$E_1^0 = \langle 100; 100 | H' | 100; 100 \rangle = \int \cdots \int d^3x_1\, d^3x_2 \frac{Z^6}{\pi^2 a_0^6} \exp\left(\frac{-2Z(r_1 + r_2)}{a_0} \right) \frac{e^2}{4\pi\varepsilon_0 r_{12}}$$

Now, to simplify this by getting rid of some constants, take

$$\frac{E_1^0}{|\Lambda_0|} = \frac{2Z^6}{\pi^2} \int \cdots \int \frac{d^3x_1\, d^3x_2}{a_0^6} \exp\left(\frac{-2Z(r_1 + r_2)}{a_0} \right) \frac{a_0}{r_{12}}, \quad (4)$$

where Λ_0 is defined above.

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To deal with $\frac{1}{r_{12}} = \frac{1}{|\vec{x}_1 - \vec{x}_2|}$, which is sometimes called the "Newtonian potential", we turn to Legendre polynomials, $P_\ell(\cos\gamma)$. By the work of the good Adrien-Marie Legendre, we know that

$$\frac{1}{r_{12}} = \frac{1}{|\vec{x}_1 - \vec{x}_2|} = \frac{1}{\sqrt{r_1^2 + r_2^2 - 2r_1 r_2 \cos\gamma}} = \sum_{\ell=0}^{\infty} \frac{r_<^\ell}{r_>^{\ell+1}} P_\ell(\cos\gamma) \qquad (5)$$

law of cosines

[Note that $\cos\gamma \equiv \cos\theta_1 \cos\theta_2 + \sin\theta_1 \sin\theta_2 \cos(\varphi_1 - \varphi_2)$]

where $r_>$ ($r_<$) is the larger (smaller) of r_1 and r_2, γ the angle between $\vec{x}_1$ and $\vec{x}_2$.
Further, recall that the addition theorem for spherical harmonics gives

$$P_\ell(\cos\gamma) = \frac{4\pi}{2\ell+1} \sum_{m=-\ell}^{\ell} Y_{\ell m}^*(\theta_1, \varphi_1) Y_{\ell m}(\theta_2, \varphi_2) \qquad (6)$$

Combining (4), (5), and (6), we get

$$\frac{E_1^\circ}{|\Lambda_0|} = \frac{2Z^6}{\pi^2 a_0^5} \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} \frac{4\pi}{2\ell+1} \iint \frac{r_<^\ell}{r_>^{\ell+1}} \exp\left(\frac{-2Z(r_1+r_2)}{a_0}\right) dr_1 dr_2 \iint d\Omega_1 Y_{\ell m}(\theta_1\varphi_1) \iint d\Omega_2 Y_{\ell m}^*(\theta_2\varphi_2)$$

$$= \frac{2Z^6}{\pi^2 a_0^5} \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} \frac{4\pi}{2\ell+1} \iint \frac{r_<^\ell}{r_>^{\ell+1}} \exp\left(\frac{-2Z(r_1+r_2)}{a_0}\right) dr_1 dr_2 \left(\sqrt{4\pi}\, \delta_{\ell,0} \delta_{m,0}\right)^2$$

$$= \frac{32Z^6}{a_0^5} \iint \frac{1}{r_>} \exp\left(\frac{-2Z(r_1+r_2)}{a_0}\right) dr_1 dr_2$$

set $R_1 = Zr_1/a_0$; $R_2 = Zr_2/a_0$

$$= 32Z \int_0^\infty dR_1 R_1^2 \left[\int_0^{R_1} dR_2 \frac{R_2^2}{R_1} e^{-2(R_1+R_2)} + \int_{R_1}^\infty dR_2 R_2 e^{-2(R_1+R_2)}\right]$$

$$= \frac{5Z}{4} = \frac{5}{2} \Rightarrow E_1^\circ = 5/2\ Ry$$

Here,

$$E_0^\circ + E_1^\circ = \left(8 - \frac{5}{2}\right)\Lambda_0 = -74.8\ \text{eV} \quad \checkmark$$